

Unit 11

GROUP 17

47 items · 114 marks

Syllabus sections in this unit

- 11.1 Physical properties of the Group 17 elements
- 11.2 The chemical properties of the halogen elements and the hydrogen halides
- 11.3 Some reactions of the halide ions
- 11.4 The reactions of chlorine

Unit 11: Group 17**1. M/J 26/P21/Q3(a)**

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (a) (i) Write an equation for reaction 1.

[1]

- (ii) Identify the reactant that behaves as a Brønsted–Lowry base in reaction 1.

[1]

2. M/J 26/P21/Q3(b)-(c)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (b)** State the formulae of the species responsible for the brown vapour in reaction 2 and for the purple vapour in reaction 3.

brown vapour

purple vapour

[1]

- (c)** Explain why reaction 3 produces purple vapour.

[1]

3. M/J 26/P21/Q3(e)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

Context: part (b)

- (b)** State the formulae of the species responsible for the brown vapour in reaction 2 and for the purple vapour in reaction 3.

brown vapour

purple vapour

[1]

- (e)** Explain why reaction 3 produces H₂S but reaction 2 does **not** produce H₂S.

[1]

4. M/J 26/P21/Q3(f)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (f) (i) Describe what is observed when Ag⁺(aq), followed by NH₃(aq), are added to a sample of NaI(aq).

observation after addition of Ag⁺(aq)

observation after addition of NH₃(aq)

[2]

- (ii) Write an ionic equation for the reaction of Ag⁺(aq) with NaI(aq).

[1]

5. M/J 26/P22/Q3(b)-(d)

Context

- 3** Separate samples of NaCl and NaBr are reacted with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	• effervescence • steamy fumes that turn blue litmus paper red
2	NaBr	• effervescence • steamy fumes initially that turn blue litmus paper red • brown vapour • pungent smelling gas

Context: part (a)(i)

- (a) (i)** Write an equation for reaction 1.

[1]

- (b)** Identify the formula of each species responsible for the observations in reaction 2.

steamy fumes

brown vapour

pungent smelling gas

[3]

- (c)** Explain why reaction 2 produces brown vapour.

[1]

- (d)** Explain why a gas with a pungent smell is **not** produced in reaction 1 but is produced in reaction 2.

[1]

6. M/J 26/P23/Q1(b)

Context

- 1** The chlorides of Period 3 elements show different physical and chemical properties.

- (b)** The reaction of chlorine with cold water is used in water purification.

- (i)** Write an equation for the reaction of chlorine with cold water.

[1]

- (ii)** Explain the use of chlorine in water purification.

[1]

7. M/J 26/P23/Q1(c)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.

(c) The relative strength of Cl_2 , Br_2 and I_2 as oxidising agents can be compared by adding NaBr(aq) to separate samples of Cl_2 and I_2 .

(i) Describe the relative reactivity of Cl_2 , Br_2 and I_2 as oxidising agents.

[1]

(ii) Complete Table 1.2 to describe the reactions of NaBr(aq) with separate samples of $\text{Cl}_2(\text{aq})$ and $\text{I}_2(\text{aq})$.

Table 1.2

aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	halide ions present in the mixture after the reaction
$\text{Cl}_2(\text{aq})$	pale green		
$\text{I}_2(\text{aq})$	red–brown		

[3]

8. M/J 26/P23/Q1(d)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.

(d) Describe what is observed when $\text{Ag}^+(\text{aq})$, followed by $\text{NH}_3(\text{aq})$, are added to a sample of NaI(aq) .

observation after addition of $\text{Ag}^+(\text{aq})$

observation after addition of $\text{NH}_3(\text{aq})$

[2]

9. MAR 26/P22/Q1(d)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(d) Calcium chlorate(I), Ca(ClO)_2 , is used for water purification in the same way as sodium chlorate(I), NaClO .

Ca(ClO)_2 can be prepared by the reaction of aqueous Ca(OH)_2 with Cl_2 .

(i) Write an ionic equation for the reaction of cold dilute OH^- ions with Cl_2 .

[1]

(ii) State why NaClO and Ca(ClO)_2 are used in water purification.

[1]

10. O/N 25/P24/Q1(a)(i)

Context

- 1** The Group 17 elements are oxidising agents.

(a) (i) Explain how the Group 17 elements act as oxidising agents.

[1]

11. O/N 25/P24/Q1(a)(iii)

Context

- 1** The Group 17 elements are oxidising agents.

(iii) A student heats equal amounts of $I_2(g)$ and $H_2(g)$ in a sealed flask. The student leaves the contents to cool.

State what you would observe during the reaction.

[1]

12. O/N 25/P24/Q1(b)(i)-(ii)

Context

- 1** The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX .



The relative bond strengths of $X-X$ and $H-X$ determine the difference in enthalpy change of the two reactions.

(i) Describe and explain the difference in the $X-X$ bond strengths of Cl_2 and Br_2 .

[2]

(ii) Describe the relative thermal stabilities of HCl and HBr .

[1]

13. O/N 25/P24/Q1(c)

Context

1 The Group 17 elements are oxidising agents.

(c) Table 1.2 shows some information about reactions of NaCl, NaBr and NaI.

Table 1.2

	NaCl	NaBr	NaI
observation with $\text{Ag}^+(\text{aq})$	white precipitate		
type of reaction with concentrated H_2SO_4	acid–base	acid–base, then redox	acid–base, then redox
observations with concentrated H_2SO_4			• black solid • yellow solid • effervescence

(i) Complete Table 1.2. [4]

(ii) Suggest an identity for the species that produces each observation in the reaction of NaI with concentrated H_2SO_4 .

black solid

yellow solid

effervescence

[2]

14. M/J 25/P23/Q6(a)(ii)-(iii)

Context: part (a) (introduction)

6 (a) HOCl reacts with $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$ in an electrophilic addition reaction.(ii) Describe how Cl_2 is used to produce HOCl .

[1]

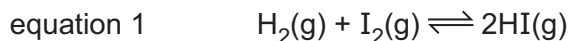
(iii) State **one** use for HOCl .

[1]

15. M/J 25/P24/Q3(d)(ii)-(iii)

Context

- 3** $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

Context: part (c)

- (c)** In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.



Use the equilibrium constant, K_c , to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = _____ mol [1]

Context: part (d) (introduction)

- (d)** The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34} .



- (ii)** Describe and explain the relative thermal stabilities of the hydrogen halides HCl and HI . [2]

- (iii)** Use the data given in **(c)** and **(d)** to suggest a value for the equilibrium constant for the **dissociation** of $\text{HBr}(\text{g})$ at 298 K.

[1]

16. MAR 25/P22/Q3(a)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (a) (i)** Complete Table 3.1. [1]

- (ii)** The bond energy values in Table 3.1 refer to the $\text{X}-\text{X}$ bond where X is the halogen.

Explain the trend in the bond strength of the $\text{X}-\text{X}$ bond in the halogens.

[2]

17. MAR 25/P22/Q3(b)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol ⁻¹	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (b)** Explain, with the use of an equation, how chlorine, Cl₂, is used in water purification. State the role of the active species produced.

[2]

18. MAR 25/P22/Q3(c)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol ⁻¹	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (c)** The sodium halides in Table 3.1 also show trends in chemical properties.

- (i)** Identify the sodium halide that reacts with concentrated H₂SO₄ to form H₂S.

[1]

- (ii)** Identify the sodium halide that **only** undergoes a Brønsted–Lowry acid–base reaction with concentrated H₂SO₄.

[1]

- (iii)** A student adds a few drops of AgNO₃(aq) to a solution of NaBr(aq). State what is observed.

[1]

19. O/N 24/P21/Q2(a)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

- (a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature. [1]
- (ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents. [1]
- (iii) State what is observed when chlorine reacts with hydrogen. [1]
- (iv) Explain why the thermal stability of the hydrogen halides decreases down the group. [1]

20. O/N 24/P21/Q2(c)(i)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (i) Identify a reagent and conditions that can be used to convert chlorine to NaClO . [1]

21. O/N 24/P23/Q2(a)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

- (a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature. [1]
- (ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents. [1]
- (iii) State what is observed when chlorine reacts with hydrogen. [1]
- (iv) Explain why the thermal stability of the hydrogen halides decreases down the group. [1]

22. O/N 24/P23/Q2(c)(i)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (i) Identify a reagent and conditions that can be used to convert chlorine to NaClO . [1]

23. M/J 24/P21/Q1(a)

- 1 (a) The elements of Group 17 are called halogens.

Complete Table 1.1.

Table 1.1

halogen	colour at 293 K
chlorine	
bromine	
iodine	

[1]

24. M/J 24/P21/Q1(b)

- (b) State the trend in volatility of the halogens chlorine, bromine and iodine. Explain your answer.

[3]

25. M/J 24/P21/Q1(c)

- (c) Iodine is made by reacting bromine with sodium iodide.

- (i) Construct an ionic equation for the reaction of bromine with sodium iodide.

[1]

- (ii) State the role of bromine in the reaction. Explain your answer.

[1]

26. M/J 24/P21/Q1(d)

- (d) Concentrated sulfuric acid is added to separate samples containing equal amounts of NaCl, NaBr and NaI. All three samples initially react to produce the hydrogen halide.

- (i) Write an equation to describe the acid–base reaction that occurs when concentrated sulfuric acid reacts with NaBr.

[1]

- (ii) Deduce which sodium halide, NaCl, NaBr or NaI, produces the largest percentage yield of hydrogen halide when concentrated sulfuric acid is added. Explain your answer by considering the relative reactivity of the halide ions as reducing agents.

identity of sodium halide

explanation

[3]

27. MAR 24/P22/Q2(b)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

- (b)** When Cl_2 reacts with **cold** NaOH(aq) , Cl_2 is both oxidised and reduced. The products are NaCl , water and **G**.

- (i)** State the type of redox reaction in which the same species is both oxidised and reduced.

[1]

- (ii)** Identify **G**.

[1]

- (iii)** Write an equation for the reaction between Cl_2 and **hot** NaOH(aq) .

[1]

- (iv)** Describe fully what is observed when $\text{AgNO}_3(\text{aq})$ is added to the aqueous solution of the chloride of sodium, followed by dilute $\text{NH}_3(\text{aq})$.

[2]

28. O/N 23/P21/Q1(c)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (c) A student does three tests on separate samples of NaCl(aq).

Complete Table 1.2 with the observations the student makes in each test.

Table 1.2

test	test	observations
1	addition of a few drops of Br ₂ (aq)	
2	addition of a few drops of concentrated H ₂ SO ₄	
3	addition of a few drops of dilute AgNO ₃ (aq)	

[3]

29. O/N 23/P23/Q1(c)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
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- (c) A student does three tests on separate samples of NaCl(aq).

Complete Table 1.2 with the observations the student makes in each test.

Table 1.2

test	test	observations
1	addition of a few drops of Br ₂ (aq)	
2	addition of a few drops of concentrated H ₂ SO ₄	
3	addition of a few drops of dilute AgNO ₃ (aq)	

[3]

30. M/J 23/P21/Q3(c)

Context

- 3** Chlorine is a very reactive element.

- (c) (i) State the reagent and conditions required for the formation of sodium chlorate(V) from Cl₂(g).

[1]

- (ii) Explain why the reaction in (c)(i) is described as a disproportionation reaction. Your answer should refer to relevant species and their oxidation numbers.

[1]

31. M/J 23/P22/Q2(d)

Context

2 Chlorine is a reactive element. It forms many compounds.

(d) (i) Write an equation for the reaction of chlorine with water.

[1]

(ii) Write an equation for the reaction of chlorine with hot NaOH(aq).

[1]

32. M/J 23/P22/Q2(e)(i)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO, to water.

(i) Identify the formula of the ion present in bleach that kills bacteria.

[1]

33. M/J 23/P23/Q4(a)

4 (a) Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

[1]

(ii) Explain why hydrogen iodide is **not** prepared by adding concentrated sulfuric acid to sodium iodide.

[2]

34. MAR 23/P22/Q2(d)(i)

Context

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(d) (i) Describe what is observed when SrI₂(aq) reacts with concentrated sulfuric acid.

[2]

35. O/N 22/P21/Q2(c)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(c) 1 cm³ of MgCl₂(aq) is placed in a test-tube. A few drops of AgNO₃(aq) are added, followed by 1 cm³ of dilute NH₃(aq).

State in full what is observed in this experiment.

[2]

36. O/N 22/P21/Q2(d)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(d) When 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$ in a test-tube, the solution remains orange.

Explain this observation.

[1]

37. O/N 22/P21/Q3(b)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(b) NaCl is one product of the reaction of chlorine gas and cold aqueous sodium hydroxide.

Identify the other products.

[1]

38. O/N 22/P22/Q2(e)(i)

Context

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(e) (i) An excess of $\text{Cl}^-(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$.

Describe what is observed. Explain your answer.

[2]

39. O/N 22/P23/Q2(c)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(c) 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is placed in a test-tube. A few drops of $\text{AgNO}_3(\text{aq})$ are added, followed by 1 cm^3 of dilute $\text{NH}_3(\text{aq})$.

State in full what is observed in this experiment.

[2]

40. O/N 22/P23/Q2(d)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(d) When 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$ in a test-tube, the solution remains orange.

Explain this observation.

[1]

41. O/N 22/P23/Q3(b)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(b) NaCl is one product of the reaction of chlorine gas and cold aqueous sodium hydroxide.

Identify the other products.

[1]

42. M/J 22/P22/Q2(d)

Context

2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(d) A sample of aqueous calcium halide, $\text{CaX}_2(\text{aq})$, contains either chloride, bromide or iodide ions.

Complete Table 2.1 to describe a two-step process that could be used to identify the halide ion present.

Table 2.1

step	method	observation with CaCl_2	observation with CaBr_2	observation with CaI_2
step 1				
step 2				

[3]

43. M/J 22/P22/Q3(d)

(d) HCl(g) is prepared by adding NaCl(s) to concentrated H_2SO_4 .

HI(g) is **not** prepared by adding NaI(s) to concentrated H_2SO_4 because the HI(g) produced also reacts with concentrated H_2SO_4 .

(i) Identify the type of reaction that occurs when NaI(s) reacts with concentrated H_2SO_4 to form HI(g) .

[1]

(ii) Write an equation for the reaction of HI(g) and concentrated H_2SO_4 .

[1]

(iii) Explain why HI(g) reacts with concentrated H_2SO_4 whereas HCl does not.

[1]

44. M/J 22/P23/Q2(b)

(b) Tennessine, Ts, is an unstable man-made element. It is found below astatine, At, in Group 17. The chemical properties of Ts and its compounds have only been predicted.

(i) Suggest an equation for the reaction of NaTs and Br_2 . Assume that Ts follows the same trends as the other elements in Group 17. Explain your answer.

equation

explanation

[2]

(ii) Some scientists predict that Ts has properties typical of metals like copper.

Complete Table 2.2 with:

- the predicted melting point of tennessine
- the lattice structure of solid chlorine, bromine and tennessine.

Assume that Ts has properties typical of metals like copper.

Table 2.2

element	chlorine	bromine	tennessine
melting point/ $^{\circ}\text{C}$	-101	-7.2	
lattice structure of crystalline solid			

[2]

45. MAR 22/P22/Q3(a)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(a) The hydrogen halides can be formed by reacting the halogens with hydrogen.

Describe and explain the relative reactivity of the halogens down the group when they react with hydrogen to form HCl , HBr and HI .

[2]

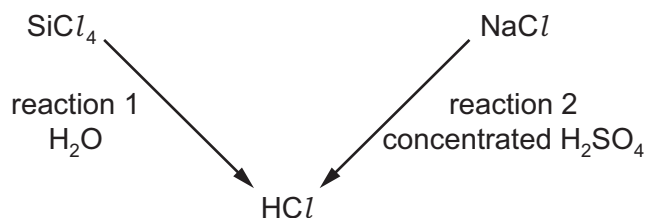
46. MAR 22/P22/Q3(b)(ii)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (b) (introduction)

(b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

**Fig. 3.1**

(ii) In reaction 2, NaCl reacts with concentrated H_2SO_4 to form HCl and NaHSO_4 . When NaBr reacts with concentrated H_2SO_4 , the products include Br_2 and SO_2 .

Identify the type(s) of reaction that occur in each case by completing Table 3.1. Explain the difference in these reactions.

Table 3.1

reactants	type(s) of reaction
NaCl and concentrated H_2SO_4	
NaBr and concentrated H_2SO_4	

explanation

.....

[3]

47. MAR 22/P22/Q3(c)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(c) When heated with a Bunsen burner, HCl does not decompose, whereas HI forms H_2 and I_2 .

Explain the difference in the effect of heating on HCl and HI .

[1]

Answers Section

1. M/J 26/P21/Q3(a)

3(a)(i)	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{NaHSO}_4 + \text{HCl}$	1	A $\text{H}_2\text{SO}_4 + 2\text{NaCl} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$ A $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ A $2\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{SO}_4^{2-}$
3(a)(ii)	$\text{Cl}^- / \text{NaCl}$	1	A sodium chloride / chloride (ion) A chlorine anion I chlorine ion I chlorine R Cl_2

2. M/J 26/P21/Q3(b)-(c)

3(b)	Brown vapour Br_2 Purple vapour I_2	1	I bromine I Br I iodine I I R iodide / bromide
3(c)	(purple vapour is produced when) HI / iodide (anions) are oxidised (by H_2SO_4) OR (sulfuric) acid / H_2SO_4 oxidises I^- (to iodine)	1	A iodide / I^- loses electrons A H_2SO_4 / S (in H_2SO_4) is reduced by HI / I^- / I in NaI / I in HI A I^- oxidised by S ((VI))(in H_2SO_4) For (VI) A 6+ / +6 / 6 I displacement I oxidation products I oxidation number unless they are specifying S(VI) in H_2SO_4 I The reaction is exothermic so the iodine / I_2 produced is vaporised R iodine is oxidised by sulfuric acid

3. M/J 26/P21/Q3(e)

3(e)	iodide / I^- is a stronger reducing agent (than bromide / Br^-) OR HBr / bromide / Br^- is a weaker reducing agent (than iodide / I^-) OR HBr / Bromide / Br^- can't reduce S(IV) / SO_2	1	A the reducing ability of the halides increases down the group A for I^- HI / NaI / I in NaI / I in HI / iodide A bromide is not strong enough to fully reduce sulfuric acid strong enough = good enough I bond energy arguments I reactivity arguments if not linked to redox I electronegativity arguments I references to H_2SO_4 being an oxidising agent I explanations
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Answer for context part (b)

3(b)	Brown vapour Br_2 Purple vapour I_2	1	I bromine I Br I iodine I I R iodide / bromide
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4. M/J 26/P21/Q3(f)

3(f)(i)	(observation after addition of $\text{Ag}^+(\text{aq})$)(pale) yellow precipitate (observation after addition of $\text{NH}_3(\text{aq})$)precipitate remains / precipitate does not dissolve / precipitate does not disappear	2	precipitate = ppt(e) = crystals = powder = (insoluble) solid M1 ignore shades R any other colour M2 A no change/no reaction/no visible change if ppt seen in M1 M2 A precipitate is insoluble M2 yellow ppt if yellow ppt seen in M1 M2 A ecf ppt remains on incorrect colour ppt in M1
3(f)(ii)	$\text{Ag}^+ + \text{I}^- \rightarrow \text{AgI}$	1	I $\text{Ag}^+ + \text{NaI} \rightarrow \text{AgI} + \text{Na}^+$ I $\text{Ag}^+ + \text{Na}^+ + \text{I}^- \rightarrow \text{AgI} + \text{I}^- + \text{Na}^+$

5. M/J 26/P22/Q3(b)-(d)

3(b)	M1 (steamy fumes) HBr M2 (brown vapour) Br ₂ M3 (pungent gas) SO ₂	3	If formula given, I name. If no formula given, I name <u>once only</u> and then A correct name. I state symbols M1 I HCl M2 I Br M3 I Br ₂ and / or HBr
3(c)	(brown vapour is made when) HBr / bromide (anions) are oxidised (by H ₂ SO ₄)	1	A H ₂ SO ₄ / S (in H ₂ SO ₄) is reduced by HBr / Br ⁻ / Br in NaBr / Br in HBr A bromide / Br ⁻ loses electrons A Br ⁻ oxidised by S (VI) (in H ₂ SO ₄) A Br ⁻ / HBr / Br in NaBr reduces S(VI) (to S(IV)) I displacement I oxidation products I oxidation numbers unless they are specifying the S(VI) in H ₂ SO ₄ R bromine is oxidised by sulfuric acid For VI allow 6+ / 6 / +6
3(d)	H ₂ SO ₄ is not (a) strong enough (oxidising agent) to oxidise Cl ⁻ / NaCl / HCl OR Cl ⁻ is not (a) strong enough (reducing agent) to reduce H ₂ SO ₄ / S (in sulfuric acid)	1	A HBr / Br ⁻ is a stronger reducing agent than HCl / Cl ⁻ A reducing ability of halides increases down the group Cl ⁻ = HCl / NaCl / Cl in NaCl / Cl in HCl / chloride Strong enough = good enough Not strong enough oxidising agent = can't oxidise I bond energy arguments I reactivity arguments if not linked to redox I electronegativity arguments I reduction product e.g. H ₂ S A H ₂ SO ₄ is (a) strong enough (oxidising agent) to oxidise Br ⁻ A Br ⁻ is (a) strong enough (reducing agent) to reduce H ₂ SO ₄ / S (in sulfuric acid) / sulfate ions Br ⁻ = HBr / NaBr / Br in NaBr / Br in HBr / bromide R HCl is a stronger oxidising agent than H ₂ SO ₄ R Cl ⁻ is a stronger oxidising agent than H ₂ SO ₄ R NaCl is a stronger oxidising agent than H ₂ SO ₄

Answer for context part (a)(i)

3(a)(i)	H ₂ SO ₄ + NaCl → NaHSO ₄ + HCl OR H ₂ SO ₄ + 2NaCl → Na ₂ SO ₄ + 2HCl	1	A Cl ⁻ + H ₂ SO ₄ → HCl + HSO ₄ ⁻ A 2Cl ⁻ + H ₂ SO ₄ → 2HCl + SO ₄ ²⁻
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6. M/J 26/P23/Q1(b)

1(b)(i)	Cl ₂ + H ₂ O → HCl + HOCl	1	A Cl ₂ + H ₂ O → 2H ⁺ + Cl ⁻ + OCl ⁻
1(b)(ii)	(HOCl AND / OR ClO ⁻) kill bacteria	1	chloric(I) acid / hypochlorous acid = HOCl ClO ⁻ = chlorate(I) I chlorine / Cl ₂ A ecf product(s) of the reaction kill bacteria if 1b(i) correct A microbes / microorganisms / microbes BOD pathogens I viruses A 'destroys' for kills I sterilises / harms / removes

7. M/J 26/P23/Q1(c)

1(c)(i)	$\text{Cl}_2 > \text{Br}_2 > \text{I}_2$ OR oxidising power decreases down the group.				1	A relative reactivity for relative oxidising power I attempted explanations												
1(c)(ii)	<table><tr><th>aqueous halogen</th><th>colour of aqueous halogen</th><th>appearance after NaBr(aq) is added</th><th>identity of halide ions present in mixture after the reaction</th></tr><tr><td>$\text{Cl}_2(\text{aq})$</td><td>pale green</td><td>orange / brown A yellow alone R red-brown</td><td>Cl^- (and Br^-)</td></tr><tr><td>$\text{I}_2(\text{aq})$</td><td>red-brown</td><td>pale(r) (red-brown) A (paler) red-brown A no change</td><td>Br^-</td></tr></table>				aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	identity of halide ions present in mixture after the reaction	$\text{Cl}_2(\text{aq})$	pale green	orange / brown A yellow alone R red-brown	Cl^- (and Br^-)	$\text{I}_2(\text{aq})$	red-brown	pale(r) (red-brown) A (paler) red-brown A no change	Br^-	3	•✓✓✓ M1 shade of brown on addition NaBr to Cl_2 must be different to that when it is added to I_2 e.g. orange in test-tube with Cl_2 AND brown in test-tube with I_2 then BOD 'brown'
aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	identity of halide ions present in mixture after the reaction															
$\text{Cl}_2(\text{aq})$	pale green	orange / brown A yellow alone R red-brown	Cl^- (and Br^-)															
$\text{I}_2(\text{aq})$	red-brown	pale(r) (red-brown) A (paler) red-brown A no change	Br^-															

8. M/J 26/P23/Q1(d)

1(d)	M1 (after addition of $\text{Ag}^+(\text{aq})$).....(pale) yellow precipitate M2 (after addition of $\text{NH}_3(\text{aq})$).....precipitate remains / precipitate does not dissolve / precipitate does not disappear	2	precipitate = ppt(e) = crystals = powder = (insoluble) solid M1 I shades R any other colour M2 A no change / no reaction / no visible change if ppt seen in M1 M2 A insoluble (if ppt given in M1) M2 A yellow ppt if yellow ppt seen in M1 M2 A precipitate is insoluble M2 A ecf ppt remains on incorrect colour ppt in M1 M2 BOD 'no observation' M2 I colour of M1 precipitate M2 I 'no reaction'
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9. MAR 26/P22/Q1(d)

1(d)(i)	$2\text{OH}^- + \text{Cl}_2 \rightarrow \text{ClO}^- + \text{Cl}^- + \text{H}_2\text{O}$	1
1(d)(ii)	(ClO^-) kills bacteria	1

10. O/N 25/P24/Q1(a)(i)

1(a)(i)	(halogens can) remove electron(s) (from a reactant)	1
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11. O/N 25/P24/Q1(a)(iii)

1(a)(iii)	purple (gas) disappears	1
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12. O/N 25/P24/Q1(b)(i)-(ii)

1(b)(i)	M1 Br—Br is weaker (than Cl—Cl) AND Br atoms are larger M2 less (significant) overlap of orbitals (in Br-Br)	2
1(b)(ii)	HCl is more thermally stable	1

13. O/N 25/P24/Q1(c)

1(c)(i)		NaCl	NaBr	NaI	4
	with Ag ⁺ (aq)	white precipitate	M1 cream / off-white ppt	M2 yellow ppt	
	with conc. H ₂ SO ₄	acid – base	acid – base, then redox	acid – base, then redox	
	observations with conc. H ₂ SO ₄	M3 steamy fumes OR fizzing	M4 brown liquid OR fizzing	- black solid - yellow solid - effervescence	
1(c)(ii)	M1 black solid = iodine / I ₂ AND yellow solid = sulfur / S _(s) M2 effervescence = hydrogen sulfide / H ₂ S(g)				2

14. M/J 25/P23/Q6(a)(ii)-(iii)

6(a)(ii)	mix / add Cl ₂ into water	1
6(a)(iii)	water purification / killing bacteria or microbes	1

15. M/J 25/P24/Q3(d)(ii)-(iii)

3(d)(ii)	M1 HCl is more stable AND (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length M2 (In HCl) stronger attraction between nucleus and shared pair of electrons OR (In HCl) (fewer shells of electrons so) less shielding OR (In HCl) greater orbital overlap so stronger interaction	2
3(d)(iii)	any value or range of values for K _c between $5.50 \times 10^{-34} < K_c < 1.26 \times 10^{-3}$	1

Answer for context part (c)

3(c)	2.46	1
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16. MAR 25/P22/Q3(a)

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	1
3(a)(ii)	- atoms larger (down the group) - distance between electron pair and nucleus greater / longer bond length / lesser orbital overlap - weaker attraction	2

17. MAR 25/P22/Q3(b)

3(b)	$\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$ ClO ⁻ / HOCl kills bacteria	2
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18. MAR 25/P22/Q3(c)

3(c)(i)	sodium iodide / NaI	1
3(c)(ii)	sodium chloride / NaCl	1
3(c)(iii)	cream(-coloured) / off-white precipitate / solid forms	1

19. O/N 24/P21/Q2(a)

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	1
2(a)(ii)	oxidising strength decreases from chlorine to iodine	1
2(a)(iii)	green colour disappears	1
2(a)(iv)	H—Hal covalent bond strength decreases down the group	1

20. O/N 24/P21/Q2(c)(i)

2(c)(i)	cold NaOH(aq)	1
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21. O/N 24/P23/Q2(a)

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	1
2(a)(ii)	oxidising strength decreases from chlorine to iodine	1
2(a)(iii)	green colour disappears	1
2(a)(iv)	H—Hal covalent bond strength decreases down the group	1

22. O/N 24/P23/Q2(c)(i)

2(c)(i)	cold NaOH(aq)	1
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23. M/J 24/P21/Q1(a)

1(a)	halogen	colour at 293 K	1
	chlorine	green	
	bromine	brown	
	iodine	(dark) grey	
All three correct for one mark			

24. M/J 24/P21/Q1(b)

1(b)	M1 (it / volatility) decreases M2 more electrons in molecules M3 (so) increased strength of instantaneous dipole – induced dipole forces (of attraction)	3
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25. M/J 24/P21/Q1(c)

1(c)(i)	$\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$	1
1(c)(ii)	(Br ₂ / bromine is an) oxidising agent AND removes electron(s) from iodide (ions) I ⁻ OR increases the oxidation number of I ⁻ / iodide (ion)	1

26. M/J 24/P21/Q1(d)

1(d)(i)	$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{HBr} + \text{NaHSO}_4$ OR $2\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow 2\text{HBr} + \text{Na}_2\text{SO}_4$	1
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1(d)(ii)	M1 sodium chloride M2 Cl^- is not a strong enough reducing agent to reduce H_2SO_4 M3 Br^- & I^- are strong enough reducing agents to react (further) with (the S in) H_2SO_4 <i>Alternative M2 and M3</i> M2 ability to behave as reducing agents increases down the group M3 EITHER Cl^- is not strong enough to reduce H_2SO_4 OR Br^- and I^- react (further) with (the S in) H_2SO_4	3
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27. MAR 24/P22/Q2(b)

2(b)(i)	disproportionation	1
2(b)(ii)	NaClO / sodium chlorate(I)	1
2(b)(iii)	$3\text{Cl}_2 + 6\text{NaOH} \rightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$	1
2(b)(iv)	white ppt / solid	1
	dissolves (in $\text{NH}_3(\text{aq})$)	1

28. O/N 23/P21/Q1(c)

1(c)	(Br_2) no visible reaction OR colourless to orange / brown / yellow solution	1
	(conc. H_2SO_4) no visible reaction OR Solution remains colourless	1
	(dil. AgNO_3) white precipitate forms	1

29. O/N 23/P23/Q1(c)

1(c)	(Br_2) no visible reaction OR colourless to orange / brown / yellow solution	1
	(conc. H_2SO_4) no visible reaction OR Solution remains colourless	1
	(dil. AgNO_3) white precipitate forms	1

30. M/J 23/P21/Q3(c)

3(c)(i)	hot aqueous sodium hydroxide / hot $\text{NaOH}(\text{aq})$	1
3(c)(ii)	oxidation number / state of <u>chlorine / Cl</u> increases / goes (from 0) to +5 AND decreases / goes (from 0) to -1	1

31. M/J 23/P22/Q2(d)

2(d)(i)	$\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HOCl} + \text{HCl}$	1
2(d)(ii)	$3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$	1

32. M/J 23/P22/Q2(e)(i)

2(e)(i)	ClO^-	1
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33. M/J 23/P23/Q4(a)

4(a)(i)	$2\text{KCl} + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{HCl}$ OR $\text{KCl} + \text{H}_2\text{SO}_4 \rightarrow \text{KHSO}_4 + \text{HCl}$	1
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4(a)(ii)	M1 HI(g) is not the only gaseous product OR SO ₂ and / or H ₂ S and / or I ₂ (g) is (also) made M2 HI / iodide (ion) is oxidised OR H ₂ SO ₄ oxidises HI OR HI reduces sulfuric acid	2
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34. MAR 23/P22/Q2(d)(i)

2(d)(i)	white precipitate yellow solid (dark) grey solid / purple gas	effervescence / misty fumes rotten egg smell	2
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35. O/N 22/P21/Q2(c)

2(c)	white ppt / solid /	1
	solid (re)dissolves (on addition of NH ₃)	1

36. O/N 22/P21/Q2(d)

2(d)	bromine / Br ₂ is not a strong enough / (too) weak as a oxidising agent (to oxidise chloride / Cl ⁻) owtte / Bromine / Br ₂ cannot oxidise chloride (ion) / Cl ⁻	1
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37. O/N 22/P21/Q3(b)

3(b)	NaClO / sodium hypochlorite / chlorate(I) AND H ₂ O / water	1
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38. O/N 22/P22/Q2(e)(i)

2(e)(i)	M1 orange colour (of solution) remains / no visible reaction	1
	M2 bromine/Br ₂ cannot oxidise chloride/Cl ⁻ OR bromine/Br ₂ is not a strong enough oxidising agent (to oxidise chloride/Cl ⁻)	1

39. O/N 22/P23/Q2(c)

2(c)	white ppt / solid /	1
	solid (re)dissolves (on addition of NH ₃)	1

40. O/N 22/P23/Q2(d)

2(d)	bromine / Br ₂ is not a strong enough / (too) weak as a oxidising agent (to oxidise chloride / Cl ⁻) owtte / Bromine / Br ₂ cannot oxidise chloride (ion) / Cl ⁻	1
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41. O/N 22/P23/Q3(b)

3(b)	NaClO / sodium hypochlorite / chlorate(I) AND H ₂ O / water	1
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42. M/J 22/P22/Q2(d)

2(d)						3
	step	method	observation with CaCl_2	observation with CaBr_2	observation with CaI_2	
	step 1	$\text{AgNO}_3(\text{aq})$ (+ $\text{HNO}_3(\text{aq})$)	white ppt	cream OR off-white ppt	(pale) yellow ppt	
	step 2	$\text{NH}_3(\text{aq})$	(ppt) dissolves / (completely) soluble (in dilute or conc) OR (forms) colourless solution	(ppt) partly soluble / slightly soluble (in dilute or conc) ALLOW (ppt) dissolves in excess (in dilute or conc) ALLOW (ppt) soluble in conc. NH_3	(ppt) insoluble / solid remains (in dilute or concentrated)	
M1 step 1 $\text{AgNO}_3(\text{aq})$ / silver nitrate solution AND step 2 $\text{NH}_3(\text{aq})$ / ammonia solution / dilute NH_3 / concentrated NH_3						
M2 ALL solid / precipitate AND correct colours described in row 1						
M3 correct observations on addition of ammonia						

43. M/J 22/P22/Q3(d)

3(d)(i)	acid-base	1
3(d)(ii)	$8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$ OR $8\text{H}^+ + 8\text{I}^- + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$	1
3(d)(iii)	<i>explanation in terms of comparison of reducing nature of HI to HCl</i> i.e. HI / I^- is stronger reducing agent compared to Cl^- / HCl OR <i>explanation in terms of additional behaviour of HI (compared to HCl)</i> i.e. HI / I^- reduces S (in sulfuric acid) / (concentrated) sulfuric acid	1

44. M/J 22/P23/Q2(b)

2(b)(i)	M1 $2\text{NaTs} + \text{Br}_2 \rightarrow 2\text{NaBr} + \text{Ts}_2$ OR $2\text{Ts}^- + \text{Br}_2 \rightarrow 2\text{Br}^- + \text{Ts}_2$ M2 Ts^- reduces Br / Br_2 OR reducing strength of halide (ion) increases down the group				2												
2(b)(ii)	<table><tr><td>element</td><td>chlorine</td><td>bromine</td><td>tennessine</td></tr><tr><td>melting point / °C</td><td>−101</td><td>−7.2</td><td>> 250</td></tr><tr><td>lattice structure of crystalline solid</td><td>simple molecular</td><td>simple molecular</td><td>giant metallic</td></tr></table>				element	chlorine	bromine	tennessine	melting point / °C	−101	−7.2	> 250	lattice structure of crystalline solid	simple molecular	simple molecular	giant metallic	2
element	chlorine	bromine	tennessine														
melting point / °C	−101	−7.2	> 250														
lattice structure of crystalline solid	simple molecular	simple molecular	giant metallic														

45. MAR 22/P22/Q3(a)

3(a)	M1: reaction less vigorous (down the group) M2: <i>Any two of the following for one mark:</i> • electronegativity decreases • less attractive to e^- addition • weaker oxidising agent • greater nuclear charge outweighing increased shielding (ENC argument)	2
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46. MAR 22/P22/Q3(b)(ii)

3(b)(ii)	M1: <i>All three correct for two marks:</i> row 1 • acid–base row 2 • acid–base • redox M2: <i>explanation</i> H_2SO_4 is strong enough to oxidise / is an oxidising agent with $\text{NaBr} / \text{HBr} / \text{bromide}$	3
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47. MAR 22/P22/Q3(c)

3(c)	H—Cl bond is stronger than H—I / BDE decreases down the group	1
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